# AMIT MANJARLY

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## TECHNICAL SKILLS

**LLM Systems:** Continuous Batching, Dynamic KV-Cache, PagedAttention, vLLM, Fine-Tuning (QLoRA, DPO), RAG
**Applied ML:** Supervised Learning, Causal Inference, Uncertainty Quantification, Conformal Prediction, PINNs, Eval
**Computer Vision:** 2D Object Detection, YOLOv8/v11, Meta SAM 2, Vision Transformers, Out-of-Distribution Detection
**Compilers & Dist.:** MIL Compilation, Apple coremltools, NVIDIA NeMo, MVCC Snapshot Isolation, Vector Clocks
**Infra & MLOps:** Docker, Kubernetes, CUDA, Linux Kernel Profiling, MLflow, DVC, FastAPI, Redis, Git CLI, CI/CD
**Languages & DBs:** Python (Advanced Systems), C++, SQL, Bash; PostgreSQL, SQLite, Apache Hive, Impala, Redis DB

## EXPERIENCE

**Resilient** Jul 2026 - Present
*AI/ML Software Engineer Intern – Arizona, USA*

- Engineered a climate risk intelligence platform using causal AI and digital twins to model extreme weather hazards.
- Built physics-informed hybrid surrogates under severe climate stress, cutting prediction RMSE by 86.7% vs a baseline.
- Applied Double ML orthogonalization to cut causal bias by 98.1%, valuing $4.46M in avoided losses across portfolios.
- Downscaled foundation climate models over complex terrains, cutting temperature RMSE by 67.3% to hit 78.0% accuracy.

**Accenture PLC** Mar 2021 - Jul 2024
*Software Engineer – Remote, India*

- Scaled enterprise Python test automation frameworks with Tricentis qTest and CI/CD, reducing manual QA cycles by 40%.
- Hardened high-volume Impala SQL database queries across 20M+ loan records, reducing pipeline data delay times by 33%.
- Governed Oracle and Hive batch ETL pipelines through Toad for Oracle, accelerating CCAR loan model throughput by 15%.
- Maintained validation rules and automated quality checks for CECL credit loss models, securing downstream reliability.
- Diagnosed Linux server workloads via SuperPuTTY to eliminate bottlenecks, driving a 25% efficiency gain across loops.
- Architected production Tableau reports for Risk Analytics Platforms to govern pipeline telemetry and portfolio health.

## OPEN-SOURCE CONTRIBUTIONS

**Apple** [GitHub] Oct 2026 - Present
*coremltools – Core ML Model Compiler & PyTorch Converter*

- Engineered PyTorch-to-CoreML conversion ops (torch.hypot, deg2rad) lowering math into MIL for Apple Neural Engine.
- Resolved dynamic type promotion and tensor broadcasting across TorchScript and TorchExport using 68 pytest suites.

**NVIDIA** [GitHub] Oct 2026 - Present
*NeMo – Conversational AI, Speech & Generative Audio Toolkit*

- Hardened score-based diffusion backbones in NVIDIA NeMo Audio with defensive conditioning checks across 17 suites.
- Automated CLI parameter validation and duration binning in NeMo ASR manifest tools, eliminating silent run errors.

## PROJECTS

**Frontier LLM Inference & Training Platform** [GitHub] Oct 2026
*PyTorch · Hugging Face · CUDA · FastAPI · Continuous Batching · QLoRA · W4A16 Quantization*

- Architected continuous-batching engine with DynamicCache recycling, slashing memory padding waste from 68.4% to 24.1%.
- Engineered SFT pipeline with ChatML prompt masking QLoRA cut peak VRAM by 79.6% (1.85 GB) on Qwen-0.5B at no loss.
- Designed zero-memory DPO reference forward pass via adapter toggling, eliminating duplicate models with +0.310 margin.

**Autonomous Driving Continual Learning (Waymo)** [GitHub] Jan 2026 - Apr 2026
*PyTorch · Waymo Open Dataset · YOLO11m · TFRecords · Uncertainty Quantification (UQ) · CUDA · MLflow*

- Attained 100% route retention and 5x faster adaptation (31.1s to 6.2s) on novel cities via continual Actor-Critic EWC.
- Trained YOLO11m on Waymo Open Dataset, securing 37.3% mAP@50 and 34.1% recall across 4 held-out AV classes.
- Deployed OOD-triggered braking in multi-head policy, preventing collisions across 20 simulated novel hazard scenarios.

## EDUCATION

**M.S. in Data Science, Analytics and Engineering** Aug 2024 - May 2026
*Arizona State University – Arizona, USA*
*Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Statistical Inference, Distributed Systems, Cloud*

**B.E. in Computer Engineering** Aug 2015 - Jun 2019
*Gujarat Technological University – Gujarat, India*
*Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, System Programming, Java*